



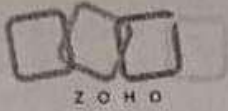
General Instructions:

- This question paper comprises 2 sections and a total of 20 questions.
- Section A comprises 10 programming questions for 40 minutes.
- Section B comprises 10 Aptitude questions for 35 minutes.
- There are no compilation errors. Ignore missing things like, #include, {}, etc.
- Incorrect answers will not result in Negative marks.
- Each question carries one mark.

Section A - Programming
10 questions | 40 minutes

Q1. What is the output for the provided input in the program below?

```
1  Input: s = "abca"
2
3  class Solution {
4      public boolean zoho(String s) {
5          int i = 0;
6          int j = s.length() - 1;
7          while(i <= j){
8              if(s.charAt(i) == s.charAt(j)){
9                  i++;
10                 j--;
11             }
12             else return help(s, i + 1, j) || help(s, i, j - 1);
13         }
14         return true;
15     }
16     public boolean help(String s, int i, int j){
17         while(i <= j){
18             if(s.charAt(i) == s.charAt(j)){
19                 i++;
20                 j--;
21             }
22             else return false;
23         }
24         return true;
25     }
26 }
```



Q2. What is the output for the provided input in the program below?

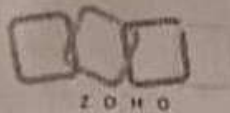
```
1  Input: nums = [6,5,4,8]
2
3  public int[] zoho(int[] nums) {
4      int[] temp = new int[101];
5      for(int i=0; i<nums.length; i++)
6          temp[nums[i]] += 1;
7      for(int j=1; j<= 100; j++)
8          temp[j] += temp[j-1];
9      for(int k=0; k< nums.length; k++) {
10         int pos = nums[k];
11         nums[k] = pos==0 ? 0 : temp[pos-1];
12     }
13     return nums;
14 }
```

Q3. What is the output for the provided input in the program below?

```
1  Input: arr = [0,3,2,1]
2
3  class Solution {
4      public boolean zoho(int[] arr) {
5          if(arr.length < 3) return false;
6          int l = 0;
7          int r = arr.length - 1;
8          while(l + 1 < arr.length - 1 && arr[l] < arr[l + 1]) l++;
9          while(r - 1 > 0 && arr[r] < arr[r - 1]) r--;
10         return l == r;
11     }
12 }
```

Q4. What is the output for the provided input in the program below?

```
1  Input: nums = [1,3,5,6], target = 7
2
3  public int zoho(int[] A, int target) {
4      int low = 0, high = A.length-1;
5      while(low<=high){
6          int mid = (low+high)/2;
```

Code: CHE230901
Duration: 75 Minutes

```
7      if(A[mid] == target) return mid;
8      else if(A[mid] > target) high = mid-1;
9      else low = mid+1;
10     }
11     return low;
12 }
```

Q5. What is the output for the provided input in the program below?

```
1  Input: arr = [1,0,0,0,1], n = 1
2
3  public class Solution {
4      public boolean zoho(int[] arr, int n) {
5          int count = 0;
6          for (int i = 0; i < arr.length; i++) {
7              if (arr[i] == 0) {
8                  boolean l = (i == 0) || (arr[i - 1] == 0);
9                  boolean r = (i == arr.length - 1) || (arr[i + 1] == 0);
10                 if (l && r) {
11                     arr[i] = 1;
12                     count++;
13                 }
14             }
15         }
16         return count >= n;
17     }
18 }
```

Q6. What is the output for the provided input in the program below?

```
1  Input: n = 5
2
3  public int[] zoho(int num) {
4      int[] f = new int[num + 1];
5      for (int i = 1; i <= num; i++) f[i] = f[i >> 1] + (i & 1);
6      return f;
7  }
```

Q7. What is the output for the provided input in the program below?

```
1  Input: arr = [3,1,7,11]
2
```



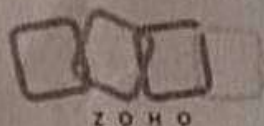
```
3 class Solution {
4     public boolean zoho(int[] arr) {
5         Arrays.sort(arr);
6         for (int i = 0; i < arr.length; i++) {
7             int target = 2 * arr[i];
8             int lo = 0, hi = arr.length - 1;
9             while (lo <= hi) {
10                 int mid = lo + (hi - lo) / 2;
11                 if (arr[mid] == target && mid != i)
12                     return true;
13                 if (arr[mid] < target)
14                     lo = mid + 1;
15                 else
16                     hi = mid - 1;
17             }
18         }
19         return false;
20     }
21 }
```

Q8. What is the output for the provided input in the program below?

```
1 Input: nums = [0,2,1,5,3,4]
2
3 public int[] buildArray(int[] nums) {
4     int[] ans = new int[nums.length];
5     for(int i = 0; i < nums.length; i++)
6         ans[i] = nums[nums[i]];
7     return ans;
8 }
```

Q9. What is the output for the provided input in the program below?

```
1 Input: str1 = "aA", str2 = "aAAbbbb"
2
3 class Solution {
4     public int zoho(String str1, String str2) {
5         int num = 0;
6         for (int i = 0; i < str2.length(); i++) {
7             if(str1.indexOf(str2.charAt(i)) != -1)
8                 num++;
9         }
10        return num;
11    }
```

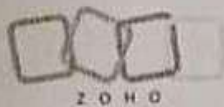
Code: CHE230901

Duration: 75 Minutes

12 }

Q10. What is the output for the provided input in the program below?

```
1  Input: nums = [2,2,3,1]
2
3  class Solution {
4      public int zoho(int[] nums) {
5          Pair<Integer, Boolean> a = new Pair<Integer, Boolean>(-1, false);
6          Pair<Integer, Boolean> b = new Pair<Integer, Boolean>(-1, false);
7          Pair<Integer, Boolean> c = new Pair<Integer, Boolean>(-1, false);
8          for (int num : nums) {
9              if ((a.getValue() && a.getKey() == num) ||
10                 (b.getValue() && b.getKey() == num) ||
11                 (c.getValue() && c.getKey() == num)) {
12                  continue;
13              }
14              if (!a.getValue() || a.getKey() <= num) {
15                  c = b;
16                  b = a;
17                  a = new Pair<Integer, Boolean>(num, true);
18              }
19              else if (!b.getValue() || b.getKey() <= num) {
20                  c = b;
21                  b = new Pair<Integer, Boolean>(num, true);
22              }
23              else if (!c.getValue() || c.getKey() <= num)
24                  c = new Pair<Integer, Boolean>(num, true);
25          }
26          if (!c.getValue())
27              return a.getKey();
28          return c.getKey();
29      }
30 }
```



Code: CHE230901
Duration: 75 Minutes

Section B - Aptitude
10 questions | 35 minutes

- Q1. Two dice are tossed. The probability that the total score is a prime number is:
- Q2. The total age of A and B is 12 years more than the total age of B and C. C is how many years younger than A?
- Q3. A man completes a journey in 10 hours. He travels the first half of the journey at the rate of 21 km/hr and the second half at the rate of 24 km/hr. Find the total journey in km.
- Q4. In how many ways can a committee, consisting of 5 men and 6 women can be formed from 8 men and 10 women?
- Q5. Three pipes A, B, and C can fill a tank in 6 hours. After working at it together for 2 hours, C is closed and A and B can fill the remaining part in 7 hours. The number of hours taken by C alone to fill the tank is:
- Q6. Two numbers A and B are such that the sum of 5% of A and 4% of B is two-third of the sum of 6% of A and 8% of B. Find the ratio of A: B.
- Q7. Mahendra Rajapaksa bought a total of 15 Aircraft, BMW, and Audi for ₹68 million. He bought more BMW than Audi. The cost of each Aircraft was ₹8 million each BMW was ₹3 million and each Audi was ₹4million. How many Aircraft did he buy?
- Q8. Two numbers are in the ratio 3: 5. If 9 is subtracted from each, the new numbers are in the ratio 12: 23. The smaller number is:
- Q9. The shopkeeper sold article A at a 20% loss and the cost price of article A was ₹4500. With that amount, he bought article B and sold it at a profit of 30%. What is the overall profit or loss in the whole transaction?
- Q10. Excluding stoppages, the speed of a bus is 54 kmph, and including stoppages, it is 45 kmph. For how many minutes does the bus stop per hour?